

Lab 10 Relations

Objective

Solving exercises from the text book.

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to

1. Understand Relations and representation of relation.
2. Understand the properties of a relation
3. Solve shorter/easier or longer / harder problems given in the textbook.

Lab Requirements

Students allowed using their lecture notes in the lab and use blackboard slides in order to solve the exercises.

Lab Assessment

- 1- Divide students to groups and let them to solve the given example.
- 2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

1. List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if
 - a) $a = b$.
 - b) $a + b = 4$.
 - c) $a > b$.
 - d) $a \mid b$.
 - e) $\gcd(a, b) = 1$.
 - f) $\text{lcm}(a, b) = 2$.
2. Represent the relation $R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$ as Arrow diagram, digraph, table and as a 0-1 matrix.
3. For each of these relations on the set $\{1, 2, 3, 4\}$, determine whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.
 - a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$
 - b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
 - c) $\{(2, 4), (4, 2)\}$
 - d) $\{(1, 2), (2, 3), (3, 4)\}$
 - e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$
4. Give an example of an irreflexive relation on the set of all people.
5. Let A be the set of students at your school and B the set of books in the school library. Let R_1 and R_2 be the relations consisting of all ordered pairs (a, b) , where student a is required to read book b in a course, and where student a has read book b , respectively. Describe the ordered pairs in each of these relations.
 - a) $R_1 \cup R_2$
 - b) $R_1 \cap R_2$
 - c) $R_1 \oplus R_2$
 - d) $R_1 - R_2$
 - e) $R_2 - R_1$
6. List the triples in the relation $\{(a, b, c) \mid a, b, \text{ and } c \text{ are integers with } 0 < a < b < c < 5\}$.

7. Let R be the relation on the set $\{0, 1, 2, 3\}$ containing the ordered pairs $(0, 1)$, $(1, 1)$, $(1, 2)$, $(2, 0)$, $(2, 2)$, and $(3, 0)$. Find the
a) reflexive closure of R . b) symmetric closure of R .
8. Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.
a) $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
b) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$
9. What is the composite of the relations R and S , where R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?
10. Find the transitive closures of these relations on $\{1, 2, 3, 4\}$.
a) $\{(1, 2), (2, 1), (2, 3), (3, 4), (4, 1)\}$
b) $\{(2, 1), (2, 3), (3, 1), (3, 4), (4, 1), (4, 3)\}$
c) $\{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$
d) $\{(1, 1), (1, 4), (2, 1), (2, 3), (3, 1), (3, 2), (3, 4), (4, 2)\}$